AXIOMETICA
AIR PLATFORM

# Autonomous Incident Response

An intelligent AIOps platform for closed-loop detection, qualification, remediation, and continuous learning.

**DETECT**
Signal intelligence

**DECIDE**
Risk-aware selection

**RESOLVE**
Autonomous remediation

**LEARN**
Continuous optimisation

## Product Overview

Version 1.2.0 | June 2026

# Axiometica AIR (Autonomous Incident Response)

### Intelligent AIOps Platform

Autonomous Incident Response, Powered by AI

# The problem with traditional NOC operations

Modern infrastructure generates thousands of alerts per day. Operators spend most of their time triaging noise, searching for runbooks, and executing repetitive remediation steps - leaving little capacity for proactive improvement or complex problem-solving.

Axiometica AIR changes this. It is an end-to-end AIOps platform that detects, qualifies, analyses, and remediates incidents autonomously - escalating to humans only when risk or novelty demands it.

**Signal detected -> classified -> qualified -> runbook selected -> executed -> verified -> summarised -> self-optimised**

**Explainable decisions**
Every step is driven by data, with clear confidence and reasoning.

**Risk-gated autonomy**
Actions are governed by CMDB context and business tolerance.

**Closed-loop learning**
Resolved incidents improve runbook selection and optimisation over time.

# 1. Signal Intelligence

## 1.1 Statistical Anomaly Detection - Sentinel Agent

Before an event becomes an incident, the Sentinel Agent evaluates it against per-service baselines built from historical telemetry. Using percentile-based statistical analysis, it classifies each signal across four tiers:

- Expected variation - suppressed
- Suspicious - monitored
- Anomalous - qualified
- Critical - immediate action

This eliminates alert fatigue at the source: only genuine signals enter the incident pipeline.

## 1.2 Intelligent Event Qualification

Qualified events are scored against configurable thresholds before a workflow is created. Events from Splunk, Datadog, Prometheus, PagerDuty, Zabbix, or native API are normalised and evaluated consistently. Low-quality signals are suppressed automatically; high-confidence signals are promoted to incidents within seconds.

## 1.3 Infrastructure Metric Correlation

The Advanced Monitoring Service moves beyond individual metric thresholds. It correlates multiple signals - for example, a CPU spike combined with an elevated syscall count - to produce composite assessments that single-metric alerts would miss. It also runs continuous external connectivity checks (HTTP, TCP, DNS, TLS), detects log error bursts and connection spikes, and feeds all signals into the incident pipeline.

# 2. Incident Intelligence

## 2.1 Multi-Factor Risk Scoring

Every incident receives a normalised 0-100 risk score computed from weighted CMDB factors. Scores normalise dynamically across active factors, ensuring incidents are ranked on the same scale even when CMDB data completeness varies. The score drives severity, priority, and approval routing automatically.

| Factor | Basis |
| --- | --- |
| User Impact | Tiered 1-5 from live user count |
| Business Criticality | Service classification from CMDB |
| Failover Availability | Full penalty when no failover exists |
| SLA Sensitivity | SLA tier from CMDB |
| Service Dependencies | Downstream dependency count |
| Historical Frequency | Past incident recurrence |

## 2.2 Event Storm Detection & Correlation

When related events fire across resources simultaneously, Axiometica AIR groups them into a single correlated storm incident rather than flooding the queue with individual tickets. The Storm Agent analyses event topology, timing, and blast radius to identify the likely root scope - suppressing child incidents and presenting operators with a unified view of the outage.

# 3. AI-Powered Response

## 3.1 AI Runbook Selection - Retrieval-Augmented Generation

When an incident qualifies, the platform uses RAG to match the incident against the runbook library. Matching considers:

- Event type and resource class
- Severity and risk profile
- Historical execution success rates
- CMDB context such as environment and service tier

The best-fit runbook is selected and queued automatically. Runbooks that succeed on similar incidents rank higher over time, allowing the system to improve with every resolution.

## 3.2 Autonomous Remediation Pipeline

For low-risk incidents with a high-confidence runbook match, the platform executes remediation without human intervention. The pipeline runs diagnostic, remediation, and verification steps, with full execution context logged for audit and review.

- Diagnostic steps confirm the root cause
- Remediation actions resolve the issue
- Verification steps confirm recovery

- Targets include Docker containers, SSH hosts, Kubernetes clusters, Azure resources, and custom action adapters

### 3.3 Intelligent Governance & Risk-Gated Approval

High-risk or high-blast-radius incidents are held for human approval before any action runs. The governance engine evaluates risk score, CMDB blast radius, and policy rules to determine routing.

- Auto-approve - low risk, known pattern, trusted runbook
- Operator approval required - elevated risk or novel scenario
- Escalation - critical systems or unresolvable conflicts

## 4. Runbook Intelligence

### 4.1 AI Runbook Generation for Novel Incidents

When no existing runbook matches, the platform generates a new runbook from scratch using an LLM, drawing on the incident's event type, CMDB context, and historical remediation patterns. Generated runbooks enter a review queue before they can be used in automated workflows.

### 4.2 AI Runbook Compose Editor

The Runbook Compose Editor gives ITOM administrators and operators an AI-assisted authoring environment for creating and editing runbooks. Operators retain full control while AI accelerates structured authoring.

- AI-generated step suggestions based on event type and target platform
- Structured diagnostic, remediation, and verification step composition
- Tool and argument configuration with contextual guidance
- Blast radius and confidence scoring for new runbooks
- One-click promotion from AI-generated draft to reviewed and enabled

### 4.3 Remediation Feedback Loop

Operators can submit outcome feedback on completed remediations. Successful runbook executions on specific incident patterns improve future ranking; failed or suboptimal outcomes adjust selection scores accordingly. The platform learns continuously from operational history in your environment.

## 5. Operational Insights

### 5.1 LLM Operational Insights

For every incident, an LLM generates a structured operational intelligence package asynchronously, off the critical execution path, so insights never delay incident response.

- Root Cause Hypothesis with confidence score and reasoning basis
- Key Concerns associated with the chosen remediation
- Remediation Rationale explaining why the selected runbook was chosen
- Historical Pattern Match with similar past incidents and outcomes
- Estimated Resolution Time
- Post-Remediation Checks for monitoring after the fix

### 5.2 Post-Resolution AI Summaries

After an incident resolves, the LLM automatically creates stakeholder-ready and audit-ready documentation.

- Executive Summary - plain-language narrative covering what happened, what was done, and the outcome
- Technical Digest - detailed event analysis, remediation reasoning, and resolution evidence

No manual write-ups. No knowledge lost between shifts.

## 6. Operator Intelligence Tools

### 6.1 AI Chat Agent

The embedded AI chat interface gives operators natural-language access to live platform intelligence. It retrieves real-time platform state - incidents, workflows, CMDB context, approval queues, and resolution history - and synthesises responses using an LLM.

- What is the current status of INC0078?
- Show me all critical incidents in the last 24 hours
- What runbook was used to resolve the last high_cpu incident on db-primary?
- Are there any incidents waiting for approval?

### 6.2 Slack Integration with Chat Agent

Axiometica AIR extends Chat Agent capability directly into operator workflows through native Slack integration.

- Critical and high severity incident alerts posted automatically to configured channels
- Storm detection alerts with affected resource summary
- Approval-required notifications with incident context
- Operators can @mention the bot in incident threads to ask follow-up questions
- Thread history is persisted across worker restarts, and Redis deduplication prevents duplicate responses in scaled deployments

## 7. Change Management AI Pipeline

Axiometica AIR extends its intelligence to proactive change management through a staged agent pipeline.

| Agent | Function |
|---|---|
| Change Risk Assessor | Scores change risk before execution begins |
| Deployment Scheduler | Calculates safe deployment windows automatically |
| Deployment Checker | Validates pre-deployment health across services |
| Deployer | Executes deployment or triggers rollback on failure |
| Deployment Verifier | Validates post-deployment state |
| Validation Agent | Runs smoke tests to confirm service health |
| Documentation Agent | Generates change records automatically on completion |

The same risk-gated governance model applies: high-risk changes require approval, while low-risk changes flow through automatically.

# 8. Platform Self-Optimisation

Axiometica AIR analyses patterns across resolved incidents and generates optimisation recommendations - suggesting threshold adjustments, risk weight changes, or approval policy tuning based on real operational data. Recommendations can be auto-applied to live configuration or queued for operator review. The platform continuously proposes improvements to its own operating parameters based on what the data shows, not intuition.

# AIOps Value Summary

| Traditional NOC | Axiometica AIR |
| --- | --- |
| Operator reads alert, creates ticket | Alert auto-qualifies and creates incident |
| Operator researches runbook manually | RAG selects best-fit runbook from history |
| Storm floods queue with individual tickets | Storm detection groups into a single correlated incident |
| Operator executes remediation steps | Pipeline executes autonomously for safe incidents |
| Risk judgement is manual and subjective | Risk score quantified from CMDB and historical data |
| Post-incident report written manually | LLM generates executive and technical summary |
| Runbook library grows slowly by hand | AI generates runbooks from novel incidents |
| Operators query dashboards to find context | Chat Agent answers in natural language |
| Alert notifications require portal login | Slack bot delivers context and accepts queries in-channel |
| Platform tuned by intuition | Self-optimisation driven by operational data |

# Confidence-Aware by Design

Every AI output in Axiometica AIR carries an explicit confidence score. Low-confidence selections route to approval. Low-confidence LLM insights are flagged with their data source so operators understand the basis for each recommendation. The platform never presents AI output as ground truth - it surfaces the reasoning behind every decision so operators can override with context.

# Built for Enterprise Operations

- Multi-source ingestion - Splunk, Datadog, Prometheus, PagerDuty, Zabbix, native API
- CMDB-driven context - ServiceNow CMDB sync with bi-directional incident mapping
- Slack integration - native bot with conversational AI in notification threads
- Full audit trail - every decision, action, and outcome logged with timestamp and actor
- Role-based access control - viewer, operator, ITOM admin, automation, and admin tiers
- Real-time updates - WebSocket-driven UI reflects pipeline state as it progresses
- LLM-agnostic - OpenAI and Anthropic Claude supported; switchable without code changes

Axiometica AIR - From alert to resolution, autonomously.

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